



AI-Powered Smart Delivery Engine for Blinkit

SwiftDidLoad 2026 — Eternal iOS Hackathon

Vision

Blinkit has perfected instant delivery.

But not every purchase needs to arrive in 10 minutes.

Every day, millions of users place small grocery orders and end up paying delivery charges that are sometimes almost equal to the product value.

At the same time, Blinkit sends multiple riders to the same neighborhood within minutes, increasing operational costs and unnecessary delivery trips.

Blinkit Wave introduces a completely new fulfillment model.

Instead of asking users,

"How fast do you want your order?"

Blinkit asks,

"When do you need it by?"

Using AI, locality-based demand prediction, intelligent delivery waves, and native iOS experiences, Blinkit automatically finds the most optimized delivery opportunity while ensuring the order arrives before the user's chosen deadline.

Problem Statement

Problem 1 — Small Orders are Expensive

Example

Need:

- Milk ₹35
- Bread ₹40

Total

₹75

Extra Charges

- Delivery Fee
- Platform Fee
- Surge Fee

Many users postpone buying essential items because delivery fees make small orders expensive.

Problem 2 — Every Order Creates a Separate Delivery

Today Blinkit processes almost every order independently.

Even if several users in the same apartment or locality order within minutes, separate delivery trips are often created.

This increases

- Rider cost
- Fuel consumption
- Traffic
- Operational expenses

Problem 3 — Users Don't Need Everything Instantly

Many groceries are needed **today**, not necessarily **right now**.

Examples

I need milk before dinner.

I need bread before 8 PM.

I need coffee tomorrow morning.

Current apps force users to either order immediately or postpone completely.

There is no intelligent flexible delivery option.

Solution

Blinkit Wave introduces one intelligent AI engine that decides the best fulfillment strategy for every order.

Instead of the user deciding how to order,

the AI decides the most efficient way to deliver it.

Core Technology — Smart Wave Engine

The **Smart Wave Engine** is the brain of Blinkit Wave.

Whenever a user places an order, the AI automatically evaluates multiple fulfillment strategies in real time.

It considers:

- Current nearby orders
- Predicted future demand
- Historical order trends
- Store capacity
- Rider availability
- Delivery radius
- Minimum order value
- User delivery deadline

The user never sees this complexity.

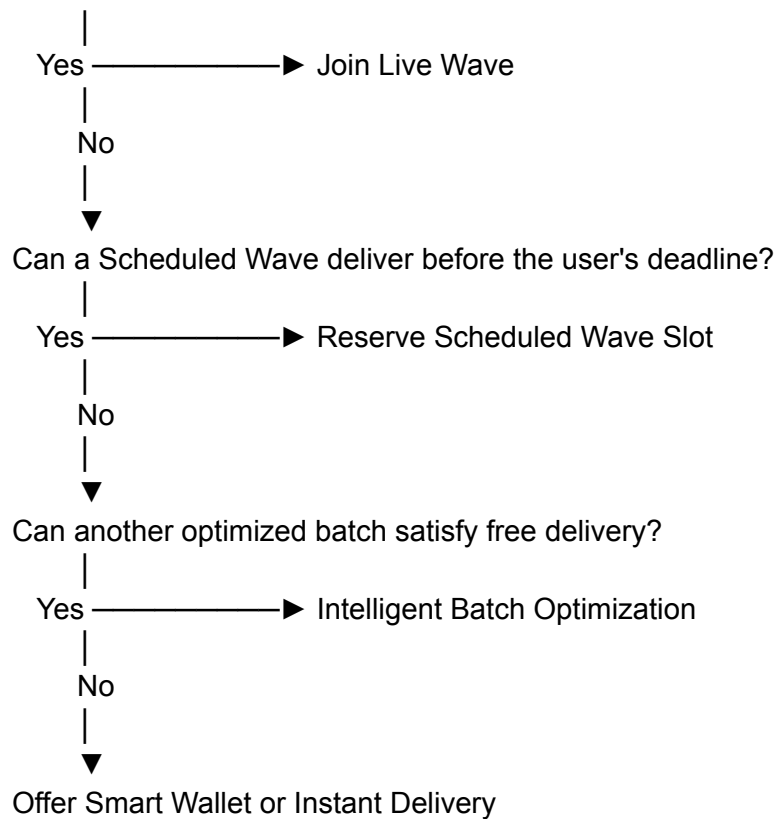
They only see the best available option.

Smart Wave Engine Decision Flow

User Places Order



Does a Live Wave exist nearby?



Mode 1 — Live Wave

Purpose

If nearby users are already placing orders, Blinkit immediately creates a Delivery Wave.

The Smart Wave Engine groups users **within a configurable delivery radius** (for example, the same apartment complex or nearby streets) while preserving complete privacy.

The user simply sees:

 Free Delivery Wave

Leaves in 3 minutes

FREE Delivery

ETA +3 min

[Join Wave]

The AI automatically checks:

- Nearby users inside the delivery radius
- Store inventory
- Rider availability
- Route optimization

If enough compatible orders exist,

the delivery becomes

✓ Free

or

Reduced Fee

depending on operational efficiency.

Intelligent Batch Optimization

This is an important enhancement.

Blinkit already has a minimum order value for free delivery.

Example

Minimum Free Delivery

₹150

Suppose

User A

₹40

User B

₹60

User C

₹55

Combined

₹155

Instead of sending three separate deliveries,

the AI recognizes that the combined value exceeds the free delivery threshold and intelligently fulfills them in a single optimized wave.

The users receive free delivery while Blinkit saves operational costs.

Even if the first two users do not reach the threshold,

the AI continues searching within the delivery radius for another compatible order until:

- the minimum order value is satisfied,
- or the delivery deadline approaches.

This entire optimization happens automatically.

Users never know about other customers.

They simply receive:

FREE Delivery unlocked through Blinkit Wave



Mode 2 — Scheduled Wave Slots

Sometimes there are not enough nearby users right now.

Instead of forcing users to pay delivery charges,

the AI predicts future demand using Blinkit's historical data.

Blinkit already knows

- morning demand
- lunch demand
- evening grocery rush
- weekend spikes
- locality shopping habits

Using these patterns,

the Smart Wave Engine creates **Wave Slots**.

Example

Sector 62

Next Wave Slot

6:30 PM

Expected Orders

26

Estimated

FREE Delivery

The user simply reserves a slot.

Deadline-Aware Slot Scheduling

This is the key innovation.

The user specifies

Need before

7 PM

The Smart Wave Engine now guarantees that every optimization happens **before that deadline**.

For example

Current Time

2 PM

Deadline

7 PM

The AI continuously evaluates every possible wave between

2 PM

↓

7 PM

The moment an optimized batch becomes possible,
the order is immediately dispatched.

It **does not wait** for the scheduled slot if enough users have already joined.

The scheduled slot is only the **latest expected dispatch time**, not a fixed departure time.

This ensures

- earlier deliveries
 - higher chances of free delivery
 - maximum operational efficiency
-

AI Decision Logic

For every order,

the AI continuously asks:

Can I create a better batch now?

↓

If YES

Dispatch immediately.

↓

If NO

Wait for the next predicted Wave Slot.

↓

Has the minimum threshold been reached?

↓

YES

Dispatch immediately.

↓

NO

Continue searching.

↓

Deadline approaching?

↓

Offer Smart Wallet or Instant Delivery.

This entire process is fully automated.



Feature 2 — Smart Wallet

Sometimes the AI cannot create an optimized delivery before the user's deadline.

Instead of forcing users to purchase unnecessary products,

Blinkit offers Smart Wallet.

Example

Need

Milk

₹30

Free Delivery Threshold

₹150

Instead of adding random snacks,

the user pays

₹150

Receives

Milk

-

₹120 Blinkit Wallet Credit

The remaining balance is automatically used in future orders.

This provides free delivery without wasteful purchases.

Feature 3 — Siri Smart Ordering

Instead of opening Blinkit,

users simply tell Siri.

Examples

"Hey Siri, order milk before 7 PM."

"Order bread before dinner."

"Buy coffee whenever a free delivery wave is available."

Blinkit immediately creates a Wave Request.

The Smart Wave Engine automatically manages the rest.

The user never has to reopen the app.

AI Prediction Inputs

The Smart Wave Engine continuously learns from:

- Previous order history
- Time of day
- Day of week
- Seasonal trends
- Festivals
- Weather
- Rider availability
- Store inventory
- Live order inflow
- Delivery density
- Locality demand

- User flexibility
 - Delivery radius
 - Minimum order value
 - User deadline
-

iOS Technologies Used

SwiftUI

- Complete application UI
 - Checkout
 - Wave cards
 - Wallet dashboard
 - Beautiful animations
-

App Intents

Allows Siri and Shortcuts to create Wave Requests.

Examples

- Order milk before 6 PM
 - Join next Wave
 - Check savings
 - Cancel Wave
-

Siri

Voice-first ordering.

Users don't need to open Blinkit.

Apple Intelligence / Foundation Models

Used for:

- Demand prediction

- Best Wave recommendation
 - Batch optimization
 - Deadline-aware decisions
 - Personalized delivery suggestions
-

ActivityKit

Live Activities show

Waiting for Wave



Wave Confirmed



Driver Assigned



1 Stop Away

Dynamic Island

Compact live updates.

Wave starts in 2 min



Free Delivery Confirmed



Driver Assigned

WidgetKit

Lock Screen

Upcoming Wave

Save ₹35

Tap to Join

Home Screen

Today's Savings

₹214

5 Waves Joined

UserNotifications

Only meaningful notifications.

Examples

- Wave Ready
 - Better Batch Found
 - Free Delivery Unlocked
 - Deadline Approaching
-

SwiftData

Stores

- Wave Requests
 - Wallet
 - User preferences
 - Delivery deadlines
 - Savings history
-

BackgroundTasks

Continuously refreshes Wave Requests, predicted slots, and delivery opportunities while respecting iOS background execution policies.

MapKit

Used internally to define **delivery radius clusters** and optimize locality-based batching without exposing user locations.

Complete User Journey

Scenario 1 — Live Wave

User needs milk immediately.



Nearby users inside the delivery radius are already ordering.



AI creates a Live Wave.



Combined order value crosses the free delivery threshold.



Order dispatched immediately.



User receives free delivery.

Scenario 2 — Scheduled Wave Slot

User needs groceries before 7 PM.



AI predicts a high-demand Wave Slot at 6:30 PM.



User reserves the slot.

↓

At 5:50 PM, enough nearby users have joined.

↓

The threshold is reached early.

↓

The order is dispatched immediately instead of waiting until 6:30 PM.

Scenario 3 — Siri

User says

"Hey Siri, order bread before dinner."

↓

Wave Request created.

↓

AI monitors Live Waves and Scheduled Wave Slots.

↓

Best batch found.

↓

Order automatically placed.

Scenario 4 — Smart Wallet

No optimized wave is found before the user's deadline.

↓

Blinkit offers

- Instant Delivery, or
- Smart Wallet.

The user chooses the option they prefer.

Business Impact

Users

- Free or reduced delivery fees
- No unnecessary shopping
- Flexible ordering
- Deadline-aware deliveries
- Voice-first ordering
- Apple-native experience

Blinkit

- Higher small-order conversion
- Lower cost per delivery
- Better rider utilization
- Improved demand forecasting
- Higher route density
- Reduced operational expenses

Environment

- Fewer rider trips
- Lower traffic congestion
- Reduced carbon emissions

One-Line Pitch

Blinkit Wave is an AI-powered Smart Delivery Engine that automatically batches nearby orders within a locality, predicts future demand through Wave Slots, guarantees delivery before the user's chosen deadline, and continuously optimizes fulfillment to maximize free delivery while minimizing operational costs.

One final suggestion (I think this is important)

I would **stop changing the core idea now**.

At this point, the idea has evolved from a simple "group orders" concept into a **full AI fulfillment engine**. Any more additions risk making it harder to explain.

From here, I'd spend all remaining time on:

1. A polished SwiftUI interface.
2. Smooth Live Activities and Dynamic Island interactions.
3. A 3-minute demo with one clear story.
4. A crisp pitch centered on the **Smart Wave Engine**.

That's what will make this stand out at SwiftDidLoad.